

Mixed ionic-electronic conduit of Ca^{2+} confined by two-dimensional multicomponent boron-carbon layers

Yang Lv, Jianfu Li,* Zhaobin Zhang, Yong Liu, Jianan Yuan, Jiani Lin, and Xiaoli Wang,*

School of Physics and Electronic Information, Yantai University, Yantai 264005, China

*Authors to whom correspondence should be addressed:

jianfuli@ytu.edu.cn and xlwang@ytu.edu.cn

Abstract

Mixed ionic-electronic conductor (MIEC) materials have widespread applications in fuel cells, all-solid-state battery electrode interface modification materials, and memory devices. Here, we reveal the two-dimensional superionic conduction between graphene-like B-C layers in the Ca-B-C ($\text{Ca}_2\text{BC}_{11}$, CaBC_5 , CaB_2C_4 , and CaB_3C_3) compounds. Through molecular dynamics simulations driven by high-precision machine learning potentials, which can be tuned within the range of 900-2100 K with defect concentrations as high as 4.8%. We discovered that the vacancy mechanism between the hexagonal prism drives its superionic diffusion. The ratio of B and C within the framework can modify the covalent (ICOHP) and ionic (Bader) interactions between the compound and Ca^{2+} , along with the interlayer spacing, thereby significantly influencing the superionic transition temperature. Furthermore, the superionic behavior across multiple phases obviates the need for synthesizing pure-phase materials. With excellent thermal and mechanical stability, these compounds of varying compositions are prospective candidates for high-temperature-resistant electrodes and interface enhancement materials and possess potential applications in many extreme conditions.

Keywords: Mixed ionic–electronic conductors (MIECs); Solid electrolyte; Crystal defects; Ionic conductivity

1. Introduction

Mixed ionic-electronic conductor (MIEC) materials, which possess both electronic and ionic conductivity, have significant applications in solid oxide fuel cells[1], hydrogen

production[2], electrochromic devices[3], neuromorphic computing[4], and phase change memory devices[5,6]. Traditional organic ionic batteries suffer from risks of explosion and leakage, making rechargeable solid-state batteries (SSBs) the most promising candidates for next-generation energy storage devices[7,8]. However, new advancements bring new challenges. Solid electrolytes (SE) typically have limited mechanical stress relief capabilities and can expand or contract during charging and discharging, leading to SE fracture, corrosion, or poor electronic or ionic contact, which can cause short circuits or increased charge transfer resistance[9]. Incorporating an intermediate layer of MIEC material between the SE and electrodes can significantly improve interface properties[9,10]. Furthermore, MIEC materials can be directly used as electrode materials in conjunction with appropriate electrolytes to enhance the electrochemical activity and stability of the interface[11–13].

Ionic conductivity is a crucial performance metric for MIEC materials. Transition metal oxides with perovskite or perovskite-related structures are among the most extensively studied MIEC materials. For example, $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$ (LSM) perovskite is widely used as a primary cathode material for high-temperature (800-1000 °C) solid oxide fuel cells (SOFCs) due to its acceptable thermal expansion coefficient (TEC), electronic conductivity, and chemical stability[14,15]. In these materials, La, Sr, and Mn form an ionic framework with nearly equivalent energy sites, allowing O^{2-} ions to diffuse freely through low-barrier ionic channels within the framework. This mechanism underpins the superionic behavior.

Compared to three-dimensional and one-dimensional materials, superionic behavior in two-dimensional materials offers more potential for electrode applications[16,17]. This is because two-dimensional materials have sufficient tolerance for changes perpendicular to the plane of the layers, which do not induce intralayer stress that would compromise the stability of the framework[18]. In the current landscape of Li-ion battery electrode materials, beyond the mainstream graphite electrodes, there is growing interest in boron-carbon (BC) materials[19–22]. Two-dimensional graphene-like BC materials can be graphene sheets where boron atoms

1 replace some carbon atoms. The inclusion of boron atoms creates electron deficiencies
2 (p-type), allowing for optimal binding energy to adsorb Li atoms[20]. As a result, BC
3 electrodes exhibit impressive high energy densities (1115 mAh/g) and possess suitable
4 adsorption energies and low migration barriers, facilitating high-speed charge and
5 discharge[21,23].

6 Given the aforementioned characteristic of two-dimensional materials with
7 tolerance for changes perpendicular to the plane, B-C graphene-like layers can
8 accommodate different ions stably between the layers. Additionally, divalent alkaline
9 earth metals ($\text{Be} = 2.19 \text{ \AA}$, $\text{Mg} = 2.40 \text{ \AA}$, and $\text{Ca} = 2.70 \text{ \AA}$) have atomic radii similar to
10 those of alkali metals ($\text{Li} = 2.20 \text{ \AA}$, $\text{Na} = 2.25 \text{ \AA}$, and $\text{K} = 2.34 \text{ \AA}$) and can theoretically
11 provide double the charge carrying capacity for an equivalent number of ions[24].
12 Moreover, calcium metal possesses a lower redox potential ($\text{Ca} \approx -2.87 \text{ V}$ versus
13 standard hydrogen electrode), is the third most abundant metallic element in the Earth's
14 crust, and exhibits good environmental friendliness[25]. It also demonstrates dendrite-
15 free electrodeposition-dissolution behavior in various electrolyte systems, making the
16 study of Ca-based mixed ionic conductors a more forward-thinking choice[26,27].
17 Consequently, we investigated the ionic conductivity behavior of the already proven
18 Ca-B-C ($\text{Ca}_2\text{BC}_{11}$, CaBC_5 , CaB_2C_4 , and CaB_3C_3) compounds. Current research on Ca-
19 B-C systems primarily focuses on high-temperature superconductivity[28].
20 Experimentally, CaB_2C_4 , CaB_2C_6 , CaB_2C_2 , and $\text{Ca}_{0.6}\text{B}_{1.2}\text{C}_{4.8}$ have been
21 synthesized[29–31]. Notably, $\text{Ca}_{0.6}\text{B}_{1.2}\text{C}_{4.8}$ has been reported as an active anode
22 material for Li-ion batteries[29].

23 Thus, whether compounds of Ca-B-C ($\text{Ca}_2\text{BC}_{11}$, CaBC_5 , CaB_2C_4 , and CaB_3C_3)
24 can indeed induce superionic behavior; what the conditions are for inducing such
25 superionic behavior; and the extent of their potential for superionic conduction—are
26 questions that compel us to delve into the exploration of the superionic properties of
27 these compounds. In this work, we employ machine learning force fields (MLFFs)
28 trained "on the fly" using the ML method implemented in VASP. After the convergence
29 of the force field, molecular dynamics simulations with machine learning potentials
30 (MLMD) are conducted to investigate the atomic dynamics and superionic behavior in

the Ca-B-C ($\text{Ca}_2\text{BC}_{11}$, CaBC_5 , CaB_2C_4 , and CaB_3C_3) compound series. We then elucidate the existence and migration mechanisms of superionic behavior, exploring its migration mechanisms from the perspective of interatomic interactions. Furthermore, introducing defects, we attempt to tune the superionic transition temperature. We find that these materials exhibit excellent ionic conductivity at high temperatures, along with good thermal stability and mechanical properties, making them highly promising two-dimensional high-temperature MIEC materials.

2. Simulation Methods

We train machine learning force fields (MLFFs) "on the fly" using the ML method implemented[32–34] in Vienna Ab initio Simulation Package (VASP, ver 6.4.2)[35,36]. The electronic exchange-correlation function used the Perdew, Burke, and Ernzerhof (PBE) generalized gradient approximation (GGA)[37]. The electron-ion interactions were used utilizing the all-electron projector augmented wave (PAW) method[38]. A plane-wave cutoff energy of 700 eV and a k-point mesh with a minimum spacing of 0.2 Å were set for further geometry optimization and self-consistent calculations. The energy was considered converged when the free energy and band structure energy variations between the two steps were less than 10^{-6} eV. For the determination of superionic behavior and exploration of diffusion mechanisms, we conducted calculations using machine learning force fields (MLFF) spanning temperatures from 900 to 2300 K to conduct molecular dynamics predictions on large supercells, aiming to find the superionic behavior. On-the-fly training is based on ab-initio molecular dynamics (AIMD) simulation samples for training structural models. Where feasible, an automated assembly of a dataset is generated and utilized for MLFF generation. In each time step, the force field predicts energy, forces, and corresponding Bayesian error estimates. Simply put, if the error surpasses a predefined threshold, another ab-initio calculation is performed, and the reference energy and forces are added to the training dataset. Conversely, in the absence of such an error, the ab-initio steps are omitted, and the system is propagated through MLFF predictions. As the force field along the trajectory improves, many ab-initio steps can be circumvented, substantially accelerating MD simulations. To verify the stability of these structures, the phonon dispersions were calculated using the Phonopy code in density functional perturbation

theory[39], and Bader's quantum theory of atoms in molecules (QTAIM) analysis was performed[40]. The Integrated Crystal Orbital Overlap Population (ICOHP) was calculated using the LOBSTER software package[41,42], and the Madelung energy and chemical bonding were analyzed using VESTA[43].

3. Results and Discussion

The B-C hexagonal rings in the Ca-B-C compounds are not in a standard graphene arrangement. As shown in Figure 1(a) and (e), the conventional unit cell of $\text{Ca}_2\text{BC}_{11}$ exhibits two types of graphene-like layers. The layer spacing is 4.468 Å. The type 1 layer consists entirely of carbon atoms, while type 2 comprises units where three hexagonal rings share a central boron atom. These two types of graphene layers lead to two different B-C bond lengths (1.517 Å and 1.510 Å) and a range of C-C bond lengths (from 1.440 Å to 1.467 Å). Interestingly, there are also distinctions in the C-C bond lengths within the type 1 layer. Figure 1(b) and (f) present the conventional unit cell of CaBC_5 , where the B-C layers are similarly composed of units of type 2 where three hexagonal rings share a central boron atom, with an interlayer spacing of 4.416 Å. The B-C bond lengths (1.532 Å to 1.539 Å) and C-C bond lengths (1.457 Å to 1.461 Å) show minor variations. As shown in Figure 1(c) and (g), with an increased boron content in CaB_2C_4 , the interlayer spacing decreases to 4.366 Å, and each hexagonal ring contains two boron atoms. The B-C bond lengths (1.548 Å to 1.564 Å) and C-C bond lengths (1.463 Å to 1.465 Å) also vary slightly. Finally, Figure 1(d) and (h) depict the conventional unit cell of CaB_3C_3 , where the graphene-like layers consist of alternating boron and carbon atoms. This structure features B-C bond lengths of 1.563 Å and 1.551 Å, with an interlayer spacing of 4.400 Å. As depicted in Figure 1(i), (j), (k), and (l), within the graphene-like structure, two B-C hexagonal rings from the upper and lower layers form a hexagonal prism unit cell. In these configurations, these distinct hexagonal prism unit cells can exist in two states: the first state features the occupancy of Ca^{2+} at the center, as illustrated in the figures; the second state is devoid of ions in the center. Except for CaB_2C_4 , which has already been synthesized experimentally[30], the thermodynamic and kinetic stability of the remaining compounds has been validated in previous studies[28], as shown in Figure S1 and Figure S2.

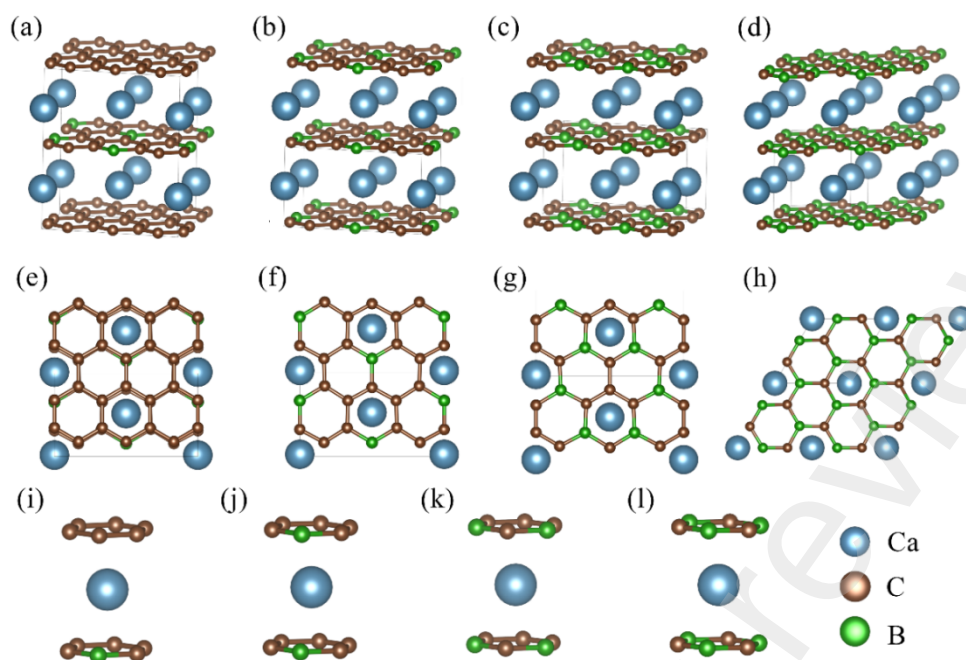


Figure 1. Overview of the structure of the compounds. (a) Structure diagram of $\text{Ca}_2\text{BC}_{11}$. (b) Structure diagram of CaBC_5 . (c) Structure diagram of CaB_2C_4 . (d) Structure diagram of CaB_3C_3 . (e) Overhead structure diagram of $\text{Ca}_2\text{BC}_{11}$. (f) Overhead structure diagram of CaBC_5 . (g) Overhead structure diagram of CaB_2C_4 . (h) Overhead structure diagram of CaB_3C_3 . (i) Diagram of the hexagonal pyramidal unit cell of $\text{Ca}_2\text{BC}_{11}$. (j) Diagram of the hexagonal pyramidal unit cell of CaBC_5 . (k) Diagram of the hexagonal pyramidal unit cell of CaB_2C_4 . (l) Diagram of the hexagonal pyramidal unit cell of CaB_3C_3 .

To demonstrate the presence of superionic behavior, we performed 60 ps molecular dynamics (MD) simulations on large supercells containing 4536 atoms (for $\text{Ca}_2\text{BC}_{11}$, CaBC_5 , and CaB_2C_4) and 4636 atoms (for CaB_3C_3) driven by MLFF. Previous studies have shown that machine-learning molecular dynamics provides very accurate predictions of the structural and vibrational properties including the complex anharmonic dynamics in the superionic materials[44]. As shown in Figure 2(a), the mean square displacement (MSD) of Ca^{2+} in $\text{Ca}_2\text{BC}_{11}$ is only $1.465 \text{ \AA}^2$ at 1400 K, increases to $4.168 \text{ \AA}^2$ at 1500 K, and further rises to $55.67 \text{ \AA}^2$ at 1900 K. In contrast, the MSD of B and C atoms at 1900 K remains low at $0.211 \text{ \AA}^2$. This indicates that at 1400 K, most Ca^{2+} are vibrating around their equilibrium positions, with only a few leaving their equilibrium sites. At 1500 K, superionic diffusion has a noticeable onset, as Ca^{2+} starts to leave their equilibrium positions more frequently. At 1900 K, Ca^{2+} diffusion becomes more pronounced and vigorous, with Ca^{2+} "flowing" like a liquid within the B-C layered framework. Meanwhile, B and C atoms only vibrate around

1 their equilibrium positions. The MSD plots for CaBC_5 , CaB_2C_4 , and CaB_3C_3 are shown
2 in Figures 2(b), 2(c), and 2(d). The trends are similar to that of $\text{Ca}_2\text{BC}_{11}$, but their
3 superionic transition temperature is higher: 1600 K for CaBC_5 , 1400 K for CaB_2C_4 , and
4 1800 K for CaB_3C_3 . Additionally, the MSD of B and C atoms remains below $0.352 \text{ \AA}^2$
5 across all temperatures, demonstrating the excellent stability of these compounds at
6 high temperatures. This may also benefit from the high-temperature stability of boron
7 carbon compounds[45,46].

8 We plotted the radial distribution function (RDF) for $\text{Ca}_2\text{BC}_{11}$, as shown in Figure
9 2(e). The RDF demonstrates that as the temperature increases, the peaks become
10 smoother, the peak values decrease, and the trough values increase. This indicates a
11 significant increase in the disorder of the Ca^{2+} within the structure. At 1900 K,
12 compared to the structure at 1400 K, the Ca^{2+} left their original occupied hexagonal
13 prism unit (Figure 1(i)). It migrated into the interior of the adjacent hexagonal prismatic
14 unit cell devoid of atomic occupancy, indicating substantial ionic diffusion. In contrast,
15 the RDF for B and C atoms, shown in Figure S3(a), indicates that the lines at
16 temperatures ranging from 1400 K to 1900K are almost entirely overlapping,
17 suggesting that the B-C framework structure remains nearly perfect at these
18 temperatures. The RDF plots for Ca^{2+} in CaBC_5 , CaB_2C_4 , and CaB_3C_3 are shown in
19 Figures 2(f), 2(g), and 2(h), respectively. These plots reveal that the Ca^{2+} exhibits
20 increasing disorder with rising temperature. Similarly, in the RDF plots for B and C
21 atoms in different compounds, shown in Figures S3(b), S3(c), and S3(d), the lines at
22 various temperatures are nearly overlapping, indicating that the B-C frameworks in
23 these compounds exhibit strong thermal stability in high temperatures.

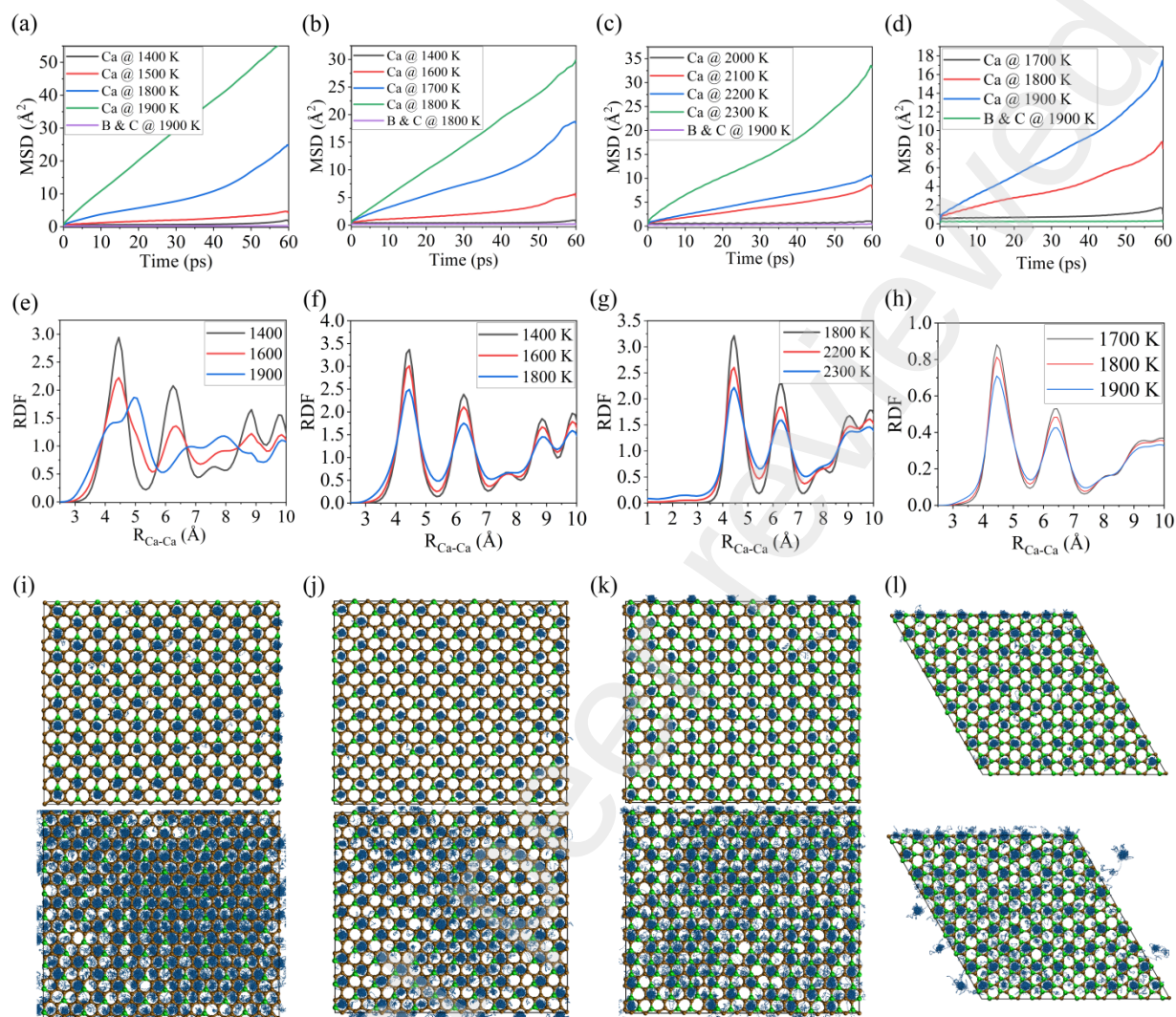
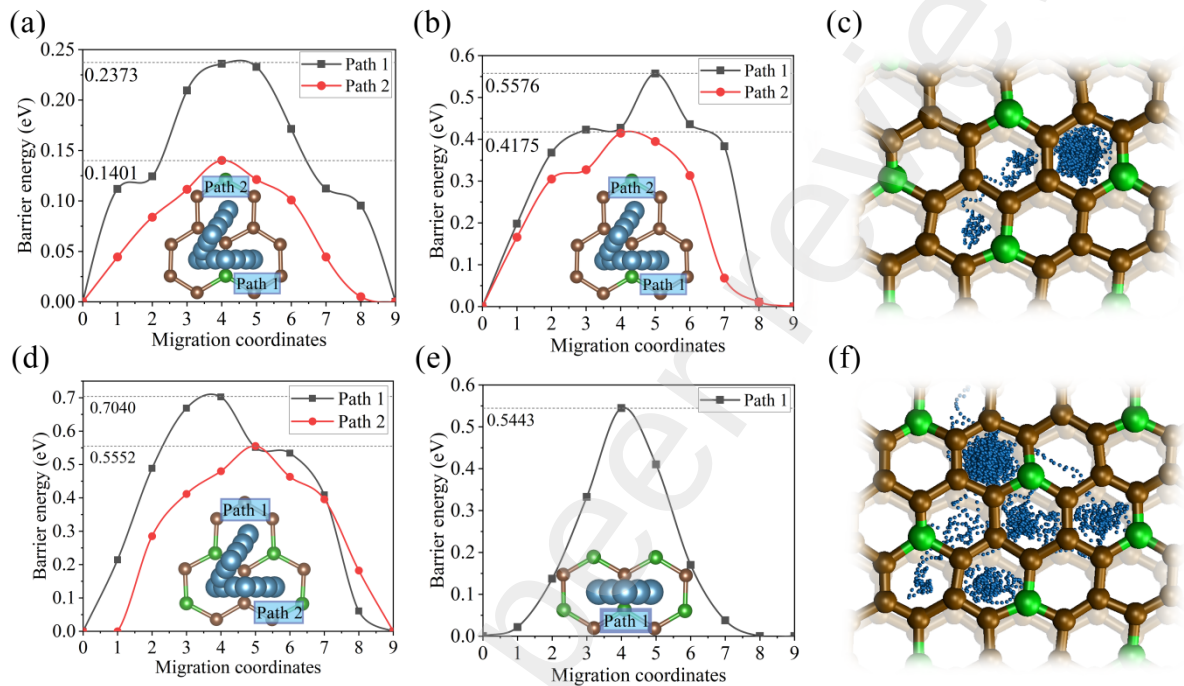


Figure 2. Characterization of superionic states in materials. MSD diagram and diffusion diagram of (a) Ca_2BC_{11} , (b) $CaBC_5$, (c) CaB_2C_4 , and (d) CaB_3C_3 . The RDF graph between Ca^{2+} and Ca^{2+} of (e) Ca_2BC_{11} , (f) $CaBC_5$, (g) CaB_2C_4 , and (h) CaB_3C_3 . Comparison of Ca cations before and after beginning to diffuse in the (i) Ca_2BC_{11} , (j) $CaBC_5$, (k) CaB_2C_4 , and (l) CaB_3C_3 .

To more intuitively demonstrate the presence of superionic behavior, we extracted the trajectories of all Ca^{2+} in a single layer of Ca_2BC_{11} at 1400 K and 1900 K over 60 ps, as shown in Figure 2(i). The upper half of the figure shows the trajectory at 1400 K, where superionic behavior is not yet apparent. It is evident that, over the 60 ps timespan, most Ca^{2+} are vibrating around their equilibrium positions, with only a few Ca^{2+} crossing the chemical bonds between C atoms to migrate to adjacent hexagonal prismatic unit cells. In contrast, the lower half of the figure shows the trajectory at 1900 K, where Ca^{2+} exhibits pronounced superionic diffusion, freely flowing like a "liquid" among equivalent sites (hexagonal prismatic unit cells). Figure 2 (j), (k), and (l) show

1 the single-layer Ca^{2+} trajectories for CaBC_5 , CaB_2C_4 , and CaB_3C_3 . At superionic
 2 transition temperature, Ca^{2+} in all these compounds easily leaves their original positions
 3 and migrates among the equivalent sites. This further confirms the superionic behavior
 4 across these materials. Notably, the superionic behavior capable of spanning multiple
 5 phases can alleviate the challenge of synthesizing pure-phase materials, which holds
 6 great promise for practical applications.



7
 8 **Figure 3.** Superionic diffusion mechanism. NEB Migration barrier diagram of (a) $\text{Ca}_2\text{BC}_{11}$, (b)
 9 CaBC_5 , (d) CaB_2C_4 , and (e) CaB_3C_3 . (c) Migration trajectories of Ca^{2+} at 1400 K in $\text{Ca}_2\text{BC}_{11}$.
 10 (f) Migration trajectories of Ca^{2+} at 1900 K in $\text{Ca}_2\text{BC}_{11}$.

11 With the full demonstration of superionic behavior in these Ca-B-C compounds,
 12 we further investigated the migration mechanism. In all Ca-B-C compounds, Ca^{2+}
 13 resides within hexagonal prism unit cells framed by two B-C hexagonal rings, as shown
 14 in Figure 1 (i), (j), (k), and (l). The top-view diagrams in Figure 1 (e), (f), (g), and (h)
 15 illustrate that each hexagonal prism unit cell containing a Ca^{2+} is surrounded by a ring
 16 of vacant hexagonal prism unit cells. We extracted the trajectory of a single Ca^{2+} in
 17 $\text{Ca}_2\text{BC}_{11}$ at 1500 K during the MLFF simulation, as shown in Figure 3(c). Ca^{2+} tends to
 18 cross the C-C bonds and move to the next hexagonal prism cells, indicating a vacancy
 19 migration mechanism. Similarly, Ca^{2+} migration in the other compounds also follows
 20 this mechanism. Furthermore, we observed that at superionic transition temperatures
 21 (1500 K for $\text{Ca}_2\text{BC}_{11}$, 1600 K for CaBC_5 , 2100 K for CaB_2C_4 , and 1800 K for CaB_3C_3),
 22 Ca^{2+} prefer to migrate through ion channels that cross C-C bonds to reach the next site,

as illustrated in Figure 3(c). At higher temperatures, Ca^{2+} diffuses through ion channels involving B-C bonds, as shown in Figure 3(f). Based on the above observations, we hypothesize that superionic behavior at lower temperatures is primarily achieved through Ca^{2+} ions crossing C-C bonds. The difference in the energy required for Ca^{2+} ions to cross C-C bonds in different structures is the main reason for the different superionic transition temperatures of these structures.

Subsequently, to verify the above hypothesis, we employed the Nudged Elastic Band (NEB) method[47,48] to compute the migration barriers of Ca^{2+} along two pathways: Path 1 involving B-C bonds and Path 2 involving only C-C bonds, as depicted in Figures 3(a), (b), (d), and (e). The graphs show that, except for CaB_3C_3 which lacks Path 2, the migration barriers along Path 1 for the remaining three compounds are notably higher than those along Path 2. This indicates that Ca^{2+} needs to overcome higher barrier energy when diffusing via Path 1. By comparing the energy barriers across these four compounds, it is observed that $\text{Ca}_2\text{BC}_{11}$, exhibiting the lowest superionic diffusion temperature (1500 K), has the lowest migration barrier along Path 2 at 0.1401 eV. Following this, CaBC_5 , with a diffusion temperature of 1600 K, shows a migration barrier along Path 2 of 0.4175 eV. CaB_3C_3 , which diffuses at 1800 K, exhibits a migration barrier of 0.5440 eV, whereas CaB_2C_4 , with the highest migration barrier of 0.5552 eV and a diffusion temperature of 2100 K, indicates a clear correspondence between diffusion barriers and the occurrence of superionic diffusion behavior in these compounds. Among the Ca-B-C ternary compounds studied, the B content in the B-C layers ranges from 8.33% to 50%. Previous conclusions suggested that higher proportions of B in graphene-like structures would lead to a greater electron deficiency, thereby increasing the adsorption energy of metal cations and adversely affecting their dynamic behavior[20]. However, our observations reveal an anomalous finding: CaB_3C_3 , with a B content of 50%, exhibits a lower temperature (1800 K) for superionic transition temperature compared to CaB_2C_4 , which has a B content of 16.67% and a higher temperature (2100 K). This discrepancy prompts further exploration into the superionic behavior of these compounds from a deeper perspective.

Therefore, to explain the aforementioned anomalous phenomenon, we further investigated the physical mechanisms of superionic behavior in these compounds from the perspective of atomic interactions. Regarding covalent interactions, our calculations of the Electron Localization Function (ELF), depicted in Figure 4(a), reveal significant electron localization between B and C atoms forming the hexagonal rings. Furthermore,

as shown in Table 1, our Integrated Crystal Orbital Hamilton Population (ICOHP) calculations demonstrate pronounced covalent interactions with ICOHP values exceeding 9.00 and 6.58 for C-C and B-C pairs, respectively. The ICOHP values between Ca and B, as well as Ca and C atoms, are of particular interest, as detailed in Table 1. These values gradually increase as the B content rises from 8.33% to 33.33%. However, upon reaching 50% B content, the ICOHP values between Ca and both B and C atoms decrease. This indicates a weakening of the covalent interactions between Ca^{2+} and the framework, facilitating easier liberation of Ca^{2+} from the constraints imposed by B and C atoms, thereby enabling diffusion to neighboring sites. Additionally, we observed that the interlayer spacing is 4.366 Å for CaB_2C_4 and increases to 4.400 Å for CaB_3C_3 , as described. This broader ion channel in CaB_3C_3 compared to CaB_2C_4 facilitates more favorable conditions for ion migration. Together, these two factors explain why CaB_3C_3 exhibits a lower superionic transition temperature compared to CaB_2C_4 . In short, previous studies suggest that alkaline earth metal Ca^{2+} , compared to alkali metal ions, can carry more charge during diffusion, albeit requiring overcoming greater ion interactions during migration[25,49]. However, this challenge can be partially alleviated through atomic or structural adjustments, such as increasing the interlayer spacing of the layered framework.

Table 1. ICOHP values between different bonds in Ca-B-C compounds.

Selected element	$\text{Ca}_2\text{BC}_{11}$	CaBC_5	CaB_2C_4	CaB_3C_3
B-C	-6.5838	-6.8498	-7.1663	-7.3487
C-C	-9.1919	-9.2764	-9.0002	-
Ca-B	-0.2839	-0.2900	-0.3245	-0.2862
Ca-C	-0.1371	-0.1658	-0.1769	-0.1721

In terms of ion interactions, we computed the Bader charge, as illustrated in Figure 4(b). As the B content increases, there is a slight increase in electron depletion by Ca^{2+} , which aligns well with previous findings. Concurrently, C atoms increase notably in electron acquisition, indicating enhanced ion interactions within the compound. The bond strength of the B-C graphene-like framework strengthens with higher B content. Subsequently, as shown in Figure 4(c), we further substantiated these observations by employing three charge transfer methods—Bader, Mulliken, and Löwdin—to

1 calculate the Madelung energy.

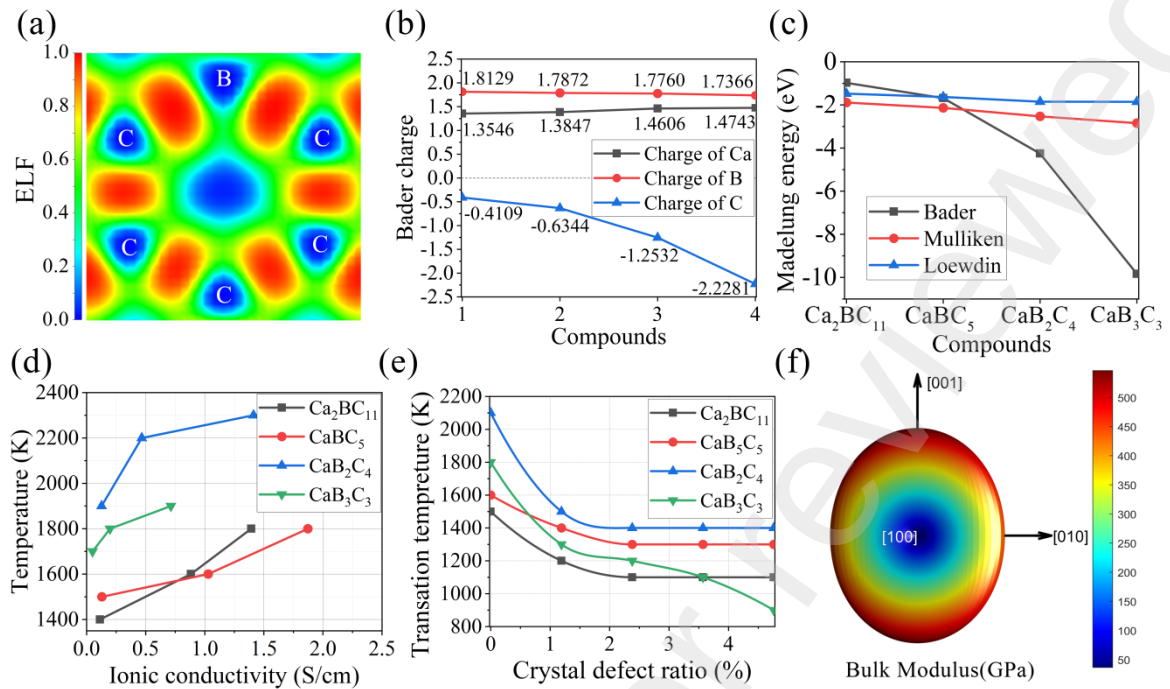


Figure 4. Physical property analysis diagram of compounds. (a) ELF diagram of B-C framework in $\text{Ca}_2\text{BC}_{11}$. (b) Bader charge transfer diagram of compounds. (c) Madelung energy diagram of compounds. (d) Ionic conductivity diagrams of different compounds. (e) Crystal defect - temperature diagram of the appearance of superionic phenomena. (f) Bulk Modulus of $\text{Ca}_2\text{BC}_{11}$.

Strong covalent and ion interactions often imply excellent mechanical properties. As shown in Figure 4(f), the materials exhibit bulk moduli exceeding 500 GPa along the [001] and [010] directions, the bulk modulus in the [100] direction also reaches 50GPa, underscoring the exceptional thermal stability and high hardness of the B-C framework. Subsequently, we calculated the ion conductivity of these compounds under different temperatures, as depicted in Figure 4(d). Research on Ca^{2+} ion conductivity is still nascent, primarily focusing on metal borohydrides and zirconium phosphate compounds. For instance, no ion conductivity was observed for $\text{CaB}_{12}\text{H}_{12}$ at 1000 K[50], and the ion conductivity of $\text{CaZr}_4(\text{PO}_4)_6$ at 1073 K is only 1.9×10^{-5} S/cm[51], which is not very promising. In contrast, although at slightly higher temperatures, the compounds in this work exhibit ion conductivities that are several orders of magnitude higher. Specifically, at 1400 K, $\text{Ca}_2\text{BC}_{11}$ exhibits an ion conductivity of 0.114 S/cm, while CaBC_5 at 1500 K shows 0.129 S/cm, CaB_2C_4 at 1900 K displays 0.126 S/cm, and CaB_3C_3 at 1800 K demonstrates 0.050 S/cm.

Furthermore, as temperature increases, the ion conductivity values exceed 0.75 S/cm. Combined with their metallic nature (Figure S4), these findings indicate that $\text{Ca}_2\text{BC}_{11}$, CaBC_5 , and CaB_3C_3 can function as MIECs. Meanwhile, CaB_2C_4 can try to develop as a superionic conductor.

Finally, we introduced crystal defects to attempt to modulate the superionic behavior of the compounds. Concerning existing materials[52–54], we set the maximum crystal point defect to 4.8% for the study. Introducing crystal defects essentially means introducing more vacancies, aiming to minimize the interactions between Ca^{2+} further to explore the potential of the materials for superionic behavior. As shown in Figure 4(e), all compounds are susceptible to the introduction of defects. With the gradual increase of defects, the temperatures at which significant superionic behavior occurs notably decrease. However, there are differences observed: $\text{Ca}_2\text{BC}_{11}$, CaBC_5 , and CaB_2C_4 show that after defects reach 2.5%, the temperatures of extensive diffusion no longer decrease with further defect increase, indicating vacancy saturation. In contrast, CaB_3C_3 does not exhibit this phenomenon. With the gradually increasing vacancies, the temperature continues to decrease, under a maximum crystal defect level of 4.8%, the superionic transition temperature of CaB_3C_3 significantly decreased by 50%, from 1800K to 900K, indicating that CaB_3C_3 is more sensitive to the introduction of defects.

4. Conclusion

In this study, we conducted an in-depth investigation into the ion conduction behavior of the ternary Ca-B-C compounds $\text{Ca}_2\text{BC}_{11}$, CaBC_5 , CaB_2C_4 , and CaB_3C_3 using MLFFs-accelerated molecular dynamics (MD). Compared with previous studies, the ionic conductivity of these structures has an advantage of several orders of magnitude. Subsequently, exception of CaB_3C_3 , from 1500 K to 2100 K, the superionic transition temperature rises with the B content within the structure. Analysis of the ICOHP, Bader, and Madelung energy and phase structures of the compounds revealed that the smaller covalent interaction between Ca^{2+} ions and the framework and the larger interlayer spacing in CaB_3C_3 are the causes for its independence from the aforementioned rule. Furthermore, by introducing crystal defects we explored the potential of these materials for superionic behavior. Remarkably, CaB_3C_3 demonstrated high sensitivity to defect introduction, with the introducing crystal defects decreasing from 1800 K to 900 K, the decline was as high as 50% when defects were introduced up to 4.8%. This result can

1 provide valuable guidance for designing and improving existing MIEC and superionic
2 conductor materials and offer a new perspective instructive for superionic behavior.
3 The superionic phenomena spanning multiple phases can significantly mitigate the
4 challenges of phase synthesis for materials and possess considerable potential for
5 practical applications.

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12 **6. Declaration of Generative AI and AI-assisted technologies in the writing process**

13 Statement: During the preparation of this work the authors used Chat GPT in order to
14 improve readability and language. After using this tool, the authors reviewed and edited
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- [1] S.K. Burnwal, S. Bharadwaj, P. Kistaiah, Review on MIEC Cathode Materials for Solid Oxide Fuel Cells, *J Mol Eng Mater* 04 (2016) 1630001. <https://doi.org/10.1142/s2251237316300011>.
- [2] W. Li, X. Zhu, Z. Cao, W. Wang, W. Yang, Mixed ionic-electronic conducting (MIEC) membranes for hydrogen production from water splitting, *Int J Hydrogen Energy* 40 (2015) 3452–3461. <https://doi.org/10.1016/j.ijhydene.2014.10.080>.
- [3] F. Huang, H. Wu, Y. Cao, Water/alcohol soluble conjugated polymers as highly efficient electron transporting/injection layer in optoelectronic devices, *Chem Soc Rev* 39 (2010) 2500–2521. <https://doi.org/10.1039/b907991m>.
- [4] Y. Van De Burgt, E. Lubberman, E.J. Fuller, S.T. Keene, G.C. Faria, S. Agarwal, M.J. Marinella, A. Alec Talin, A. Salleo, A non-volatile organic electrochemical device as a low-voltage artificial synapse for neuromorphic computing, *Nat Mater* 16 (2017) 414–418. <https://doi.org/10.1038/NMAT4856>.
- [5] K. Gopalakrishnan, R. S. Shenoy, C. T. Rettner, K. Virwani, D. S. Bethune, R. M. Shelby, G. W. Burr, A. Kellock, R. S. King, K. Nguyen, A. N. Bowers, M. Jurich, B. Jackson, A. M. Friz, T. Topuria, P. M. Rice, B. N. Kurdi, Highly-Scalable Novel Access Device based on Mixed Ionic Electronic Conduction(MIEC) Materials for High Density Phase Change Memory (PCM) Arrays, in: 2010 Symposium on VLSI Technology Digest of Technical Papers, 2010: pp. 205–206.
- [6] G. W. Burr, K. Virwani, R. S. Shenoy, A. Padilla, M. BrightSky, E. A. Joseph, M. Lofaro, A. J. Kellock, R. S. King, K. Nguyen, A. N. Bowers, M. Jurich, C. T. Rettner, B. Jackson, D. S. Bethune, R. M. Shelby, T. Topuria, N. Arellano, P. M. Rice, B. N. Kurdi, K.

- 1 Gopalakrishnan, Large-scale (512kbit) integration of Multilayer-ready Access-
 2 Devices based on Mixed-Ionic-Electronic-Conduction (MIEC) at 100% yield, in: 2012
 3 Symposium on VLSI Technology Digest of Technical Papers, 2010: pp. 41–42.
- 4 [7] R. Schmuch, R. Wagner, G. Hörpel, T. Placke, M. Winter, Performance and cost of
 5 materials for lithium-based rechargeable automotive batteries, *Nat Energy* 3 (2018)
 6 267–278. <https://doi.org/10.1038/s41560-018-0107-2>.
- 7 [8] S. Randau, D.A. Weber, O. Kötz, R. Koerver, P. Braun, A. Weber, E. Ivers-Tiffée, T.
 8 Adermann, J. Kulisch, W.G. Zeier, F.H. Richter, J. Janek, Benchmarking the
 9 performance of all-solid-state lithium batteries, *Nat Energy* 5 (2020) 259–270.
 10 <https://doi.org/10.1038/s41560-020-0565-1>.
- 11 [9] K. So Yeon, J. Li, Porous Mixed Ionic Electronic Conductor Interlayers for Solid-State
 12 Batteries, *Energy Material Advances* 2021 (2021).
 13 <https://doi.org/10.34133/2021/1519569>.
- 14 [10] E.J. Moon, J.K. Hong, S.K. Mohanty, M. Yang, K. Ihm, H. Lee, H.D. Yoo, Coating
 15 lithium titanate anodes with a mixed ionic-electronic conductor for high-rate lithium-
 16 ion batteries, *J Power Sources* 559 (2023).
 17 <https://doi.org/10.1016/j.jpowsour.2023.232657>.
- 18 [11] Z. Jin, X. Kong, H. Huang, Y. Jiang, W. Xiang, Y. Xu, L. Zhang, R. Peng, C. Wang,
 19 Garnet-type solid-state mixed ionic and electronic conductor, *Energy Storage Mater*
 20 59 (2023). <https://doi.org/10.1016/j.ensm.2023.102788>.
- 21 [12] I. Riess, Mixed ionic-electronic conductors-material properties and applications,
 22 *Solid State Ionic* 157 (2003) 1–17. www.elsevier.com/locate/ssi.

- [13] S. Wang, L. Zhao, Y. Dong, H. Zhu, Y. Yang, H. Xu, B. Wang, Y. Yuan, Y. Ren, X. Huang, W. Quan, Y. Li, Y. Huang, C.M. Settens, Q. He, Y. Sun, H. Wang, Z. Xiao, W. Liu, X. Xiao, R. Fu, Q. Li, Y.S. Chu, Z. Zhang, Q. Liu, A.M. Minor, J. Zhang, Z. Tang, J. Li, Pre-zeolite framework super-MIEC anodes for high-rate lithium-ion batteries, *Energy Environ Sci* 16 (2022) 241–251. <https://doi.org/10.1039/d2ee02918a>.
- [14] H. Yokokawa, N. Sakai, T. Kawada, M. Dokiya, THERMODYNAMIC ANALYSIS ON INTERFACE BETWEEN PEROVSKITE ELECTRODE AND YSZ ELECTROLYTE, *Solid State Ion* 40 (1990) 398–401.
- [15] S.P. Jiang, Development of lanthanum strontium manganite perovskite cathode materials of solid oxide fuel cells: A review, *J Mater Sci* 43 (2008) 6799–6833. <https://doi.org/10.1007/s10853-008-2966-6>.
- [16] S. Wang, J. Fu, Y. Liu, R.S. Saravanan, J. Luo, S. Deng, T.K. Sham, X. Sun, Y. Mo, Design principles for sodium superionic conductors, *Nat Commun* 14 (2023). <https://doi.org/10.1038/s41467-023-43436-3>.
- [17] K.H. Cho, P.K. Jain, Superionic Conduction in One-Dimensional Nanostructures, *ACS Nano* 16 (2022) 12445–12451. <https://doi.org/10.1021/acsnano.2c03732>.
- [18] R.P. Joshi, B. Ozdemir, V. Barone, J.E. Peralta, Hexagonal BC3: A robust electrode material for Li, Na, and K ion batteries, *Journal of Physical Chemistry Letters* 6 (2015) 2728–2732. <https://doi.org/10.1021/acs.jpclett.5b01110>.
- [19] H. Wang, Y. Li, Y. Wang, S. Hu, H. Hou, Highly microporous graphite-like BC_xO_{3-x}/C nanospheres for anode materials of lithium-ion batteries, *J Mater Chem A Mater* 5 (2017) 2835–2843. <https://doi.org/10.1039/c6ta10778h>.

- [20] D. Das, R.P. Hardikar, S.S. Han, K.R. Lee, A.K. Singh, Monolayer BC2: An ultrahigh capacity anode material for Li ion batteries, *Physical Chemistry Chemical Physics* 19 (2017) 24230–24239. <https://doi.org/10.1039/c7cp04451h>.
- [21] K. Belasfar, A. El Kenz, A. Benyoussef, First-principles study of BC7 monolayer an ultra-high capacity anode for lithium-ion and sodium-ion batteries applications, *Mater Chem Phys* 257 (2021). <https://doi.org/10.1016/j.matchemphys.2020.123751>.
- [22] D. Li, P. Dai, Y. Chen, R. Peng, Y. Sun, H. Zhou, Lithium Borocarbide LiBC as an Anode Material for Rechargeable Li-Ion Batteries, *Journal of Physical Chemistry C* 122 (2018) 18231–18236. <https://doi.org/10.1021/acs.jpcc.8b03763>.
- [23] X.J. Ye, R. Zhao, X. Xiong, X.H. Wang, C.S. Liu, A first-principles study of the BC3N2 monolayer and a BC3N2/graphene heterostructure as promising anode materials for sodium-ion batteries, *Physical Chemistry Chemical Physics* 26 (2024) 11738–11745. <https://doi.org/10.1039/d3cp04804g>.
- [24] M. Rahm, R. Hoffmann, N.W. Ashcroft, Atomic and Ionic Radii of Elements 1–96, *Chemistry - A European Journal* 22 (2016) 14625–14632. <https://doi.org/10.1002/chem.201602949>.
- [25] H. Song, J. Su, C. Wang, Hybrid Solid Electrolyte Interphases Enabled Ultralong Life Ca-Metal Batteries Working at Room Temperature, *Advanced Materials* 33 (2021). <https://doi.org/10.1002/adma.202006141>.
- [26] D. Wang, X. Gao, Y. Chen, L. Jin, C. Kuss, P.G. Bruce, Plating and stripping calcium in an organic electrolyte, *Nat Mater* 17 (2018) 16–20. <https://doi.org/10.1038/NMAT5036>.

- [27] A. Ponrouch, C. Frontera, F. Bardé, M.R. Palacín, Towards a calcium-based rechargeable battery, *Nat Mater* 15 (2016) 169–172. <https://doi.org/10.1038/nmat4462>.
- [28] C. Zhang, H. Tang, C. Pan, H. Jiang, H.-J. Sun, K.-M. Ho, C.-Z. Wang, Machine learning guided discovery of superconducting calcium borocarbides, *Phys Rev B* 108 (2023) 024512–024519.
- [29] G. Tei, R. Miyamae, A. Kano, High capacity graphite-like calcium boridecarbides as a novel anode active material for lithium-ion batteries, *Sustain Energy Fuels* 3 (2019) 1929–1936. <https://doi.org/10.1039/c9se00061e>.
- [30] M. Wörle, D. Widmer, F. Von Rohr, F. Krumeich, C. Mensing, R. Nesper, Opening a new series of calcium boridecarbides with heterographene nets by $\text{Ca}_{1-x}(\text{B,C})_6$ and $\text{Ca}_{1-x}(\text{B,C})_8$ - Two members of the $(\text{CaB}_2)_n$ compound series, *Z Anorg Allg Chem* 636 (2010) 1447–1453. <https://doi.org/10.1002/zaac.201000134>.
- [31] B. Albert, K. Schmitt, CaB_2C_2 : Reinvestigation of a semiconducting boride carbide with a layered structure and an interesting boron/carbon ordering scheme, *Inorg Chem* 38 (1999) 6159–6163. <https://doi.org/10.1021/ic9907821>.
- [32] R. Jinnouchi, F. Karsai, G. Kresse, On-the-fly machine learning force field generation: Application to melting points, *Phys Rev B* 100 (2019). <https://doi.org/10.1103/PhysRevB.100.014105>.
- [33] R. Jinnouchi, J. Lahnsteiner, F. Karsai, G. Kresse, M. Bokdam, Phase Transitions of Hybrid Perovskites Simulated by Machine-Learning Force Fields Trained on the Fly with Bayesian Inference, *Phys Rev Lett* 122 (2019).

- 1 <https://doi.org/10.1103/PhysRevLett.122.225701>.
- 2 [34] R. Jinnouchi, F. Karsai, C. Verdi, R. Asahi, G. Kresse, Descriptors representing two-
3 and three-body atomic distributions and their effects on the accuracy of machine-
4 learned inter-atomic potentials, *Journal of Chemical Physics* 152 (2020).
5 <https://doi.org/10.1063/5.0009491>.
- 6 [35] G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy
7 calculations using a plane-wave basis set, *Phys Rev B* 54 (1996) 11169–11186.
- 8 [36] J. Hafner, Ab-initio simulations of materials using VASP: Density-functional theory
9 and beyond, *J Comput Chem* 29 (2008) 2044–2078.
10 <https://doi.org/10.1002/jcc.21057>.
- 11 [37] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized Gradient Approximation Made
12 Simple, *Phys Rev Lett* 77 (1996) 3865–3868.
- 13 [38] J.P. Perdew, Y. Wang, Accurate and simple analytic representation of the electron-
14 gas correlation energy, *Phys Rev B* 45 (1992) 15–1992.
- 15 [39] A. Togo, F. Oba, I. Tanaka, First-principles calculations of the ferroelastic transition
16 between rutile-type and CaCl₂-type SiO₂ at high pressures, *Phys Rev B Condens
17 Matter Mater Phys* 78 (2008). <https://doi.org/10.1103/PhysRevB.78.134106>.
- 18 [40] W. Tang, E. Sanville, G. Henkelman, A grid-based Bader analysis algorithm without
19 lattice bias, *Journal of Physics Condensed Matter* 21 (2009).
20 <https://doi.org/10.1088/0953-8984/21/8/084204>.
- 21 [41] V.L. Deringer, A.L. Tchougréeff, R. Dronskowski, Crystal orbital Hamilton population
22 (COHP) analysis as projected from plane-wave basis sets, *Journal of Physical*

- 1 Chemistry A 115 (2011) 5461–5466. <https://doi.org/10.1021/jp202489s>.
- 2 [42] S. Maintz, V.L. Deringer, A.L. Tchougréeff, R. Dronskowski, LOBSTER: A tool to
3 extract chemical bonding from plane-wave based DFT, J Comput Chem 37 (2016)
4 1030–1035. <https://doi.org/10.1002/jcc.24300>.
- 5 [43] K. Momma, F. Izumi, VESTA 3 for three-dimensional visualization of crystal,
6 volumetric and morphology data, J Appl Crystallogr 44 (2011) 1272–1276.
7 <https://doi.org/10.1107/S0021889811038970>.
- 8 [44] T. Miyagawa, N. Krishnan, M. Grumet, C.R. Baecker, W. Kaiser, D.A. Egger, Accurate
9 Description of Ion Migration in Solid-State Ion Conductors from Machine-Learning
10 Molecular Dynamics, (2024). <http://arxiv.org/abs/2401.11244>.
- 11 [45] D. Fan, S. Lu, Y. Guo, X. Hu, Two-dimensional stoichiometric boron carbides with
12 unexpected chemical bonding and promising electronic properties, J Mater Chem C
13 Mater 6 (2018) 1651–1658. <https://doi.org/10.1039/c7tc04505k>.
- 14 [46] V. Domnich, S. Reynaud, R.A. Haber, M. Chhowalla, Boron carbide: Structure,
15 properties, and stability under stress, Journal of the American Ceramic Society 94
16 (2011) 3605–3628. <https://doi.org/10.1111/j.1551-2916.2011.04865.x>.
- 17 [47] G. Henkelman, H. Jónsson, Improved tangent estimate in the nudged elastic band
18 method for finding minimum energy paths and saddle points, Journal of Chemical
19 Physics 113 (2000) 9978–9985. <https://doi.org/10.1063/1.1323224>.
- 20 [48] D. Sheppard, R. Terrell, G. Henkelman, Optimization methods for finding minimum
21 energy paths, Journal of Chemical Physics 128 (2008).
22 <https://doi.org/10.1063/1.2841941>.

- [49] M.E. Arroyo-De Dompablo, A. Ponrouch, P. Johansson, M.R. Palacín, Achievements, Challenges, and Prospects of Calcium Batteries, *Chem Rev* 120 (2020) 6331–6357. <https://doi.org/10.1021/acs.chemrev.9b00339>.
- [50] Z. Lu, F. Ciucci, Metal Borohydrides as Electrolytes for Solid-State Li, Na, Mg, and Ca Batteries: A First-Principles Study, *Chemistry of Materials* 29 (2017) 9308–9319. <https://doi.org/10.1021/acs.chemmater.7b03284>.
- [51] N. Katsuhiko, I. Shoichiro, I. Kaname, E. Hisahiko, Framework Structure, Phase Transition, and Transport Properties in $MZr(PO_4)_6$ Compounds ($M=Mg, Ca, Sr, Ba, Mn, Co, Ni, Zn, Cd, \text{ and } Pb$), *The Chemical Society of Japan* 65 (1992) 3221–3227.
- [52] F. Xiong, S. Tan, X. Yao, Q. An, L. Mai, Crystal defect modulation in cathode materials for non-lithium ion batteries: Progress and challenges, *Materials Today* 45 (2021) 169–190. <https://doi.org/10.1016/j.mattod.2020.12.002>.
- [53] Y. Xu, M. Zhou, X. Wang, C. Wang, L. Liang, F. Grote, M. Wu, Y. Mi, Y. Lei, Enhancement of Sodium Ion Battery Performance Enabled by Oxygen Vacancies, *Angewandte Chemie* 127 (2015) 8892–8895. <https://doi.org/10.1002/ange.201503477>.
- [54] C. Ma, J. Alvarado, J. Xu, R.J. Clément, M. Kodur, W. Tong, C.P. Grey, Y.S. Meng, Exploring Oxygen Activity in the High Energy P2-Type $Na_{0.78}Ni_{0.23}Mn_{0.69}O_2$ Cathode Material for Na-Ion Batteries, *J Am Chem Soc* 139 (2017) 4835–4845. <https://doi.org/10.1021/jacs.7b00164>.